

Novelty Statement

The proposed Brain-Controlled Verbal Communication Enabled Wheelchair introduces a unique combination of brain-computer interface technology, obstacle detection, and verbal communication in a single low-cost assistive system.

Although individual technologies such as EEG signal processing, motorized wheelchairs, and obstacle detection have been explored independently in previous research, existing systems are often expensive, computationally intensive, or dependent on cloud connectivity.

Our work differs in the following aspects:

Novel Features

- Integration of EEG-based eye blink control with wheelchair navigation.
- Combined mobility assistance and verbal communication in one device.
- Low-cost hardware architecture suitable for developing countries.
- Modular design enabling future upgrades.
- Safety-oriented obstacle detection during navigation.
- Real-time response with minimal computational requirements.

Original Contribution

The complete system architecture, hardware integration, software implementation, control logic, testing methodology, and user interaction workflow have been independently designed and developed by our team. Existing literature has been used only for academic reference and benchmarking purposes.